Mathieu **Néron**

Staff Software Engineer

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Summary

Staff software engineer at CodeRabbit, with 15+ years in production systems. At Highnote, I owned risk and sanctions-compliance systems for a card platform that settles **$650M+ a month** over **8M+ card transactions**, **50+ programs** and **1.6M active cards**. Before that, I was Senior → LMTS at Tableau, on authoring and server features used by **87K+ customer companies** and **3M+ Tableau Public creators**. I also build applied-ML projects (**Medea**: a multi-modal PyTorch classifier with RAG explanations, F1 0.90 on channel-grouped CV).

Experience

CodeRabbit — *San Francisco, CA*

Staff Software Engineer — *Nov. 2026 – Present*

Highnote — *San Francisco, CA*

Senior Software Engineer — *Jul. 2023 – Nov. 2026*

- *Java, Spring, GraphQL, gRPC, GCP (Spanner, BigQuery, Pub/Sub), Temporal, Elasticsearch, Redis, Spring AI*
- Owned **sanctions screening** (OFAC, EU/UN) end to end and built its matching engine (phonetic index, Jaro-Winkler scoring). Moved the true/false-positive lookup off the money-movement hot path, which replaced a Redis cache I had built. Cut daily database writes by **76%** and made 12-hour runs resume from checkpoints.
- Sole engineer on **FinCEN 314(a)** screening, the statutory law-enforcement search that partner banks must answer. Designed the Temporal pipeline from FTPS ingest to encrypted delivery of results. Live since Sep 2024 against **4.4M+ account holders**; I made it generic and onboarded a second bank.
- Built the **money-movement risk feature store**: BigQuery features materialized into Redis, plus real-time velocity counters for push and pull card transfers. Warm reads take **4–5 ms** against a 50 ms budget, and fraud rules for five programs run on it.
- Led the security design to move **34 services** from a custom LLM gateway to **Spring AI 2.0**. Wrote a 7-threat model for Security and Infra, then shipped the foundation behind a two-layer kill switch, with fail-closed PII scrubbing and a tenant-scoped response cache. The proof of concept removed ∼2,000 lines.
- Built **two-way SMS fraud verification** on **Temporal** and **Twilio**, so cardholders can confirm or deny a flagged transaction. Chose Twilio's Messaging API over its Conversations API on cost, because durable workflows already hold the conversation state. Live since Jan 2026; the CTO announced it publicly.
- Built an **impossible-travel fraud rule**: haversine distance between geocoded card-present transactions inside a time window. Live for **10 customers**.
- Brought **Temporal** and **Elasticsearch** into the risk domain, where both are now standard. Wrote design guides that other engineers built on, closed a cross-bank data exposure in the bank portal, and built AI investigation and code-review tooling that the team adopted.

Tableau (Salesforce) — *Palo Alto, CA*

Senior Software Engineer → LMTS — *Jun. 2019 – Jul. 2023*

- *React, Redux, TypeScript, Java, C++, Qt* — Promoted to Lead Member of Technical Staff in 2020.
- Onboarded a new AI-predictive-model content type in Tableau Server for **Salesforce Einstein Discovery** integration — contributing to the server stack delivering **80B+ AI-powered predictions per day**.
- Drove the **Object Graph** view in Tableau's Data pane from beta to GA — the default relationships editor in Tableau's multi-table data model, part of the authoring stack behind Tableau Public's **7M+ vizzes** from **3M+ creators**; shipped transition animations, tooltips, sorting, and re-rooting.
- Contributed to the **Data Preview Area** hybridization — metadata grid plus embedded relationship editor in a single surface.

Paysafe — *Montréal, Canada*

Java Developer — *2016 – 2019*

- Shipped the new Angular 2+ **Merchant Portal** frontend from greenfield.
- **Cut frontend test runtime 6**× by migrating from PhantomJS to Nightmare.

eSignLive — *Montréal, Canada*

Java Developer — *2014 – 2016*

- **Cut Windows document converter downtime by 50%**.
- Integrated Equifax, Twilio, PDFBox, and Google Noto Fonts.

Avanade — *Montréal, Canada*

Application Developer — *2012 – 2014*

- Web engineering for Canadian National Railway (Sitecore / ASP.NET) and Hydro-Québec (SAPUI5).

Projects

github.com/mathieu-neron/RealTube

Personal learning project

RealTube — crowdsourced AI-generated YouTube content filter — *2026*

- Full-stack browser extension (Chrome/Firefox, Manifest V3) for crowdsourced flagging and hiding of AI-generated YouTube content across feed, search, recommendations, and Shorts.
- Built as a hands-on Go + Python learning exercise: **Go** (Fiber v3) on the low-latency lookup path, **Python + FastAPI** for the inference-adjacent services, **TypeScript + React 19** extension, PostgreSQL + Redis + Docker + Prometheus.
- Privacy-engineered URL lookup using a **k-anonymity hash-prefix** scheme (HIBP-style) so the backend never sees full URLs in plaintext; contract-tested API and end-to-end test coverage.

github.com/mathieu-neron/medea

Personal learning project (natural follow-up to RealTube)

Medea — multi-modal AI-generated YouTube classifier — *2026*

- Built an end-to-end ML pipeline to learn the full applied-ML stack hands-on: yt-dlp ingest → four-modality feature extraction (**CLIP ViT-B/32** visual, **faster-whisper** + **wav2vec2** anti-spoofing audio, **MiniLM** text, handcrafted scalars) → **Chroma** vector DB → small **PyTorch** MLP head → FastAPI server with retrieval-augmented **local-LLM rationale** via Ollama.
- Used **channel-grouped LeaveOneGroupOut CV** (10 folds, ∼94-clip dataset) after UMAP showed channels as 10 distinct clusters — random video-level splits would have memorized channel identity, not "AI-ness." Final MLP **F1 0.90** / **ROC-AUC 0.96** on this honest split.
- Diagnosed a silent-audio failure mode (anti-spoof model misfires on no-speech clips) and fixed it with `has_speech` × `ai_voice_prob` interaction features (F1 0.87 → 0.90); added a **temporal prior cap** for pre-2017 uploads to bound MLP extrapolation outside the training cone.

Skills

Languages	Java, TypeScript/JavaScript, Python, Go, C/C++, C#
Frameworks	Spring, Spring AI, Temporal, gRPC/Protobuf, GraphQL, React/Redux, Angular, Node.js, Qt, .NET, ANTLR
Platforms	GCP (Spanner, BigQuery, Pub/Sub), AWS, Elasticsearch, Redis, PostgreSQL, Oracle, MySQL, Docker, Kubernetes, Terraform, Bazel
ML / AI	PyTorch, Hugging Face Transformers, sentence-transformers, CLIP, Whisper, Chroma, MLflow, Ollama, RAG, Vertex AI
Domain	Sanctions and AML (OFAC, FinCEN 314(a)), card fraud risk, ACH and instant payments
Spoken	French (native), English (fluent)

Education

Université de Sherbrooke

Sherbrooke, Canada

B.Eng., Computer Engineering — *2007 – 2011*

- Specialization in Software Engineering and Artificial Intelligence.

Internships

Electronic Arts Canada

Burnaby, Canada

Software Engineer Intern — *Summer 2011*

- Built a distributed C++ / C# load-test harness running thousands of concurrent bots against **SSX**'s multiplayer server; used internally and demoed at **Gamescom 2011**.

Autodesk | Ubisoft | Lockheed Martin

Canada

Earlier Internships — *2008 – 2010*

- **Autodesk** (2010) — animation plug-in inside Maya (C++ / Qt). **Ubisoft** (2009) — QA automation (C# / .NET). **Lockheed Martin** (2008) — software evaluation.